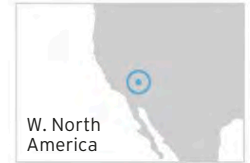


Hoover Dam

A massive structure that powered the industrial and agricultural growth of the southwestern US

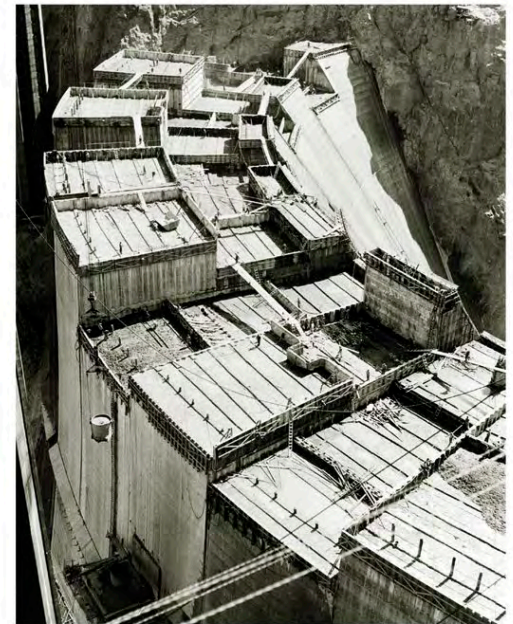


With the expansion of the southwestern economy in the 1800s and 1900s, planners began to look to the mighty Colorado River as a source of power and water for a growing population. In December 1928, President Calvin Coolidge authorized the construction of a giant dam on the borders of Nevada and Arizona.

Building the dam

To allow construction to begin, the flow of the Colorado was diverted through tunnels driven into the canyon walls. Watertight enclosures called coffer dams were built to protect the site from flooding, and the ground was cleared. The first concrete was poured in 1933, and by the time the dam was completed, it was the largest concrete structure ever made.

The challenge for the engineers was to contain 8.44 cubic miles (35.2 cubic km) of water behind the dam. Part of the resistance was provided by the weight of the concrete itself; additionally, with the point of the arch facing upstream, water was directed against the canyon's walls, serving to compress and strengthen the structure. The partially built dam—then known as the Boulder Dam—was opened by President Franklin D. Roosevelt in 1935. It was later renamed the Hoover Dam after President Herbert Hoover, who oversaw its initial construction.



△ ASSEMBLING THE SLABS

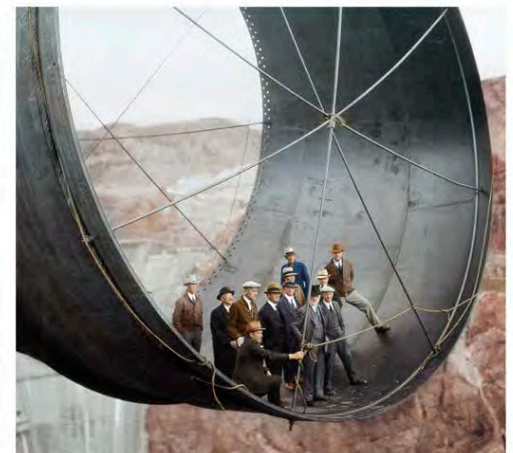
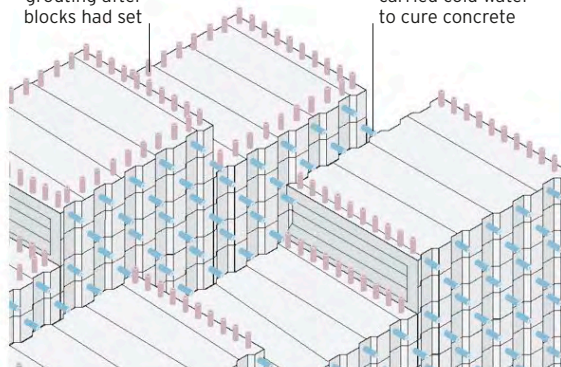
The dam was built of columns of rectangular concrete slabs (see panel) containing steel pipes through which cool river water, and then ice-cold water from a refrigeration plant, was poured to cure the concrete.

COOLING THE CONCRETE

When planning the dam, engineers calculated that if the concrete for the dam was poured continuously, it would take 125 years to cool. It was therefore poured in separate slabs, each about 50 ft (15 m) square and 5 ft (1.5 m) high.

pipes filled with grouting after blocks had set

cooling pipes carried cold water to cure concrete



△ PUBLIC APPEARANCE

This posed publicity shot taken while the dam was under construction belies the fact that 112 people lost their lives building the structure, and a further 42 died from carbon monoxide poisoning while working in the diversion tunnels.